FIITJEE

ALL INDIA TEST SERIES

OPEN TEST

JEE (Main)-2020

TEST DATE: 09-02-2020

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 75 questions.
- Each subject (PCM) has 25 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each part has only three sections: **Section-A**, **Section-B and Section-C**.

  Section-A (01 – 20, 26 – 45, 51 – 70) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 22, 46 – 47, 71 – 72) *contains 6 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

  Section-C (23 – 25, 48 – 50, 73 – 75) *contains 9 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

Physics

PART – I

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A particle is projected with speed v at an angle α to the horizontal. If the magnitude of instantaneous velocity at time 't' is equal to the magnitude of average velocity from t = 0 to t = t. Find the time 't'.

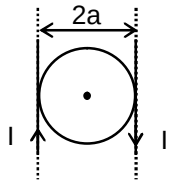
(A) $\frac{2v\sin\alpha}{3g}$

(B) $\frac{4v\sin\alpha}{3g}$

(C) $\frac{3v\sin\alpha}{4g}$

(D) $\frac{3v\sin\alpha}{2g}$

2. In the given figure two straight infinite conductors carry a current I. They are parallel and separated by a distance 2a. A circular conducting ring of radius 'a' in the plane of the parallel wires lies between the two straight conductors and is insulated from them. The mutual inductance between the circular ring and the straight conductors is



(A) $\mu_0 a$

(B) $2\mu_0 a$

(C) $\frac{\mu_0 a}{2}$

(D) $4\mu_0 a$

3. A sphere of radius r is at a potential V and is immersed in a conducting medium of conductivity σ. Find the current flowing from sphere to infinity.

(A) $2\pi\sigma rV$

(B) $4\pi\sigma rV$

(C) $\frac{\pi\sigma rV}{2}$

(D) $\frac{\pi\sigma rV}{4}$

4. A particle with charge q is travelling with a velocity v parallel to a wire with a uniform linear charge distribution λ per unit length. The wire also carries a current I. What should be the velocity of the charged particle to travel in a straight line parallel to the wire at a distance r from the wire? (c = speed of light)

(A) $\frac{2\lambda c^2}{I}$

(B) $\frac{\lambda c^2}{2I}$

(C) $\frac{\lambda c^2}{I}$

(D) $\frac{4\lambda c^2}{I}$

5. In an artificial medium, air pressure and volume are related as $P = \frac{K}{V^2}$, where K is a constant. Considering air as an ideal gas of density ρ, then the speed of sound in air is

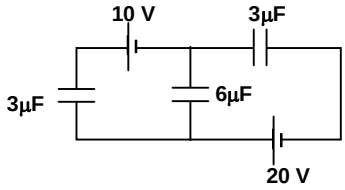
(A) $\sqrt{\frac{2P}{\rho}}$

(B) $\sqrt{\frac{P}{\rho}}$

(C) $\sqrt{\frac{P}{2\rho}}$

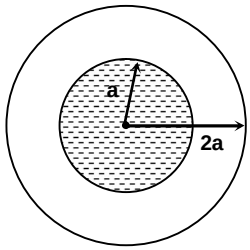
(D) $\sqrt{\frac{4P}{\rho}}$

6. In the given circuit, the charge on 6 μF capacitor in steady state is



(A) 7.5 μC
(B) 15 μC
(C) 30 μC
(D) 45 μC

7. Water is completely filled in the spherical cavity of radius 'a' of a thick spherical shell of inner and outer radii 'a' and '2a' respectively as shown. The initial temperature of water is 80°C and the surrounding temperature is θ_0 = 30°C. The thermal conductivity of the spherical shell varies as $k = \frac{\alpha}{r^2}$, where 'r' is the distance from the centre of shell and α is a constant. The time required to decrease the temperature of water from 80°C to 60°C is (ρ = density of water, S = specific heat of water)



(A) $\frac{\rho S a^4}{2\alpha} \ell n\left(\frac{5}{3}\right)$

(B) $\frac{\rho S a^4}{3\alpha} \ell n\left(\frac{5}{3}\right)$

(C) $\frac{\rho S a^4}{4\alpha} \ell n\left(\frac{5}{3}\right)$

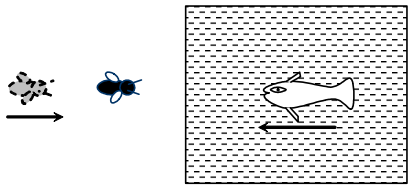
(D) $\frac{\rho S a^4}{5\alpha} \ell n\left(\frac{5}{3}\right)$

8. An ancient piece of wood shows an activity of 7.8 disintegration per second, per gram of ^{14}C. What is the age of piece of the wood if fresh ^{14}C, shows an activity of 31.2 disintegration per second per gram? Half life of ^{14}C is 5570 years.

(A) 16710 years
(B) 33420 years

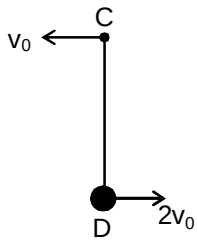
(C) 11140 years
(D) 22280 years

9. A fish in an aquarium approaching the left wall at 4 m/s observes a fly is approaching it at 8 m/s. If the refractive index of water is $\mu = \frac{4}{3}$. Find the actual velocity of the fly.



(A) 3 m/s
(B) 6 m/s
(C) 9 m/s
(D) 12 m/s

10. A small ball of mass 'm' is suspended from inextensible massless string of length l. At certain instant of time the velocity of point C is v_0 (in leftward direction) and of ball D is $2v_0$ (in rightward direction). The tension in the string at this instant of time is



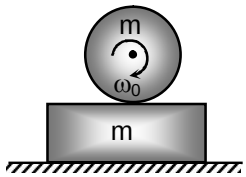
(A) $mg + \frac{4mv_0^2}{\ell}$

(B) $mg - \frac{9mv_0^2}{\ell}$

(C) $mg + \frac{mv_0^2}{\ell}$

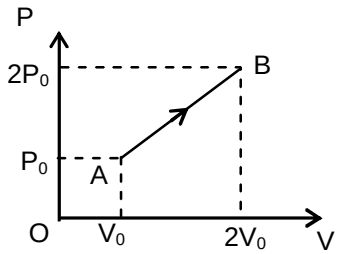
(D) $mg + \frac{9mv_0^2}{\ell}$

11. A solid cylinder of mass m = 5 kg and radius R = 0.4 m is given an initial angular velocity ω_0 = 40 rad/s about a horizontal axis through its centre and gently placed on a plank of mass m = 5 kg. The coefficient of friction between the cylinder and plank is μ = 0.2. The plank rests on a smooth horizontal surface. The time after which pure rolling motion of cylinder will start relative to the plank is (Take g = 10 m/s^2)



(A) 2 sec
(B) 3 sec
(C) 4 sec
(D) 5 sec

12. An ideal diatomic gas is taken through a process AB as shown in the P-V diagram. The molar heat capacity of the gas in this process will be :



(A) 3R
(B) 4R
(C) $\frac{9R}{2}$
(D) $\frac{R}{2}$

13. The electric field strength produced by the radiations coming from 200 W bulb at a 5 m distance is E. The electric field strength produced by the radiations coming from 100 W bulb at the same distance 5m is

(A) $\frac{E}{2}$
(B) 2E

(C) $\frac{E}{\sqrt{2}}$

(D) $\sqrt{2}E$

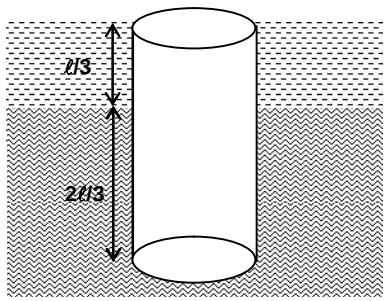
14. A particle 'A' of mass 'm' moving with an initial velocity 'v' collides elastically with a particle B of mass 2m which is initially at rest. The ratio of the de-Broglie wavelengths $\left(\frac{\lambda_A}{\lambda_B}\right)$ of the particles A and B after collision is

(A) 2
(B) 3
(C) 8
(D) 4

15. In YDSE of equal width slits, if the intensity at the center of the screen is I_0, then the intensity at a distance of $\frac{\beta}{4}$ from the central maxima is (β is fringe width, d is separation between the slits, λ is wavelength of light, D is separation between the plane of slits and the screen) (D>>d and d>>λ)

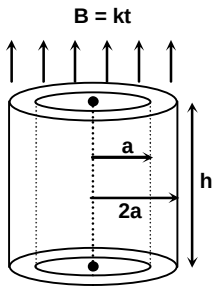
(A) I_0
(B) $\frac{I_0}{2}$
(C) $\frac{I_0}{3}$
(D) $\frac{I_0}{4}$

16. A cylinder of length ℓ floats with its entire length immersed in two liquids as shown in the figure. The upper liquid is water and lower liquid has density twice that of water. The two liquids are immiscible. The cylinder is in equilibrium. Find the specific gravity of the material of the cylinder.



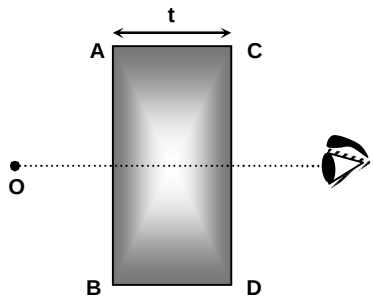
(A) 2
(B) 5
(C) 4/3
(D) 5/3

17. A ring of circular cross section is made of a material of resistivity 'ρ'. The inner and outer radii of ring are a and 2a respectively and its thickness is 'h'. The ring is placed in a time varying magnetic field B = kt (where k is a constant) which is parallel to the axis of the ring as shown in the figure. The induced current in the ring is



(A) $\frac{3kha^2}{4\rho}$
(B) $\frac{3kha^2}{2\rho}$
(C) $\frac{5kha^2}{4\rho}$
(D) $\frac{5kha^2}{2\rho}$

18. A glass slab is placed between an object O and observer. The refractive index of glass slab varies with distance x from the face AB as $\mu = \mu_0\left(1+\frac{x}{t}\right)$, where μ_0 = 1. How much closer (compared to original distance) the object appears to the observer. (Consider normal incidence only)



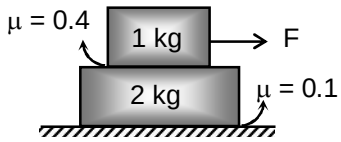
(A) t(1 + ℓn2)

(B) t(1 – ℓn2)

(C) t(2 + ℓn2)

(D) t(2 – ℓn2)

19. In the figure shown, the friction coefficient between the block of mass 1 kg and the plank of mass 2 kg is 0.4 while that between the plank and horizontal floor is 0.1. For what value of F will the block move with double the acceleration of that of the plank? (Take g = 10 m/s^2)



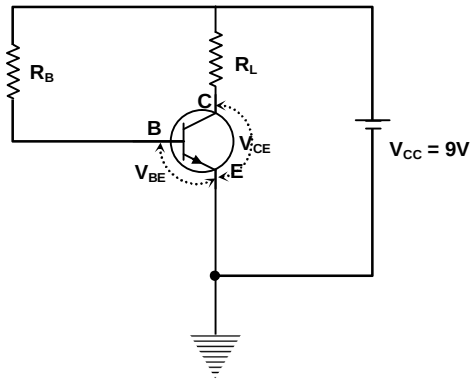
(A) 5 N

(B) 6 N

(C) 10 N

(D) 12.5 N

20. An n-p-n transistor is connected in common emitter mode as shown in the figure. The terminal of a 9 V battery is connected to the collector through a load resistance R_L = 1 kΩ and to the base through a resistance R_B. If the base-emitter voltage V_{BE} = 1.4 V, the collector-emitter voltage V_{CE} = 5 V and β = 100. Then the value of resistance R_B is



(A) 130 kΩ

(B) 160 kΩ

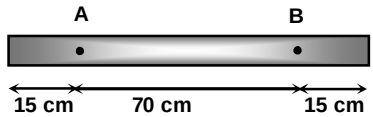
(C) 190 kΩ

(D) 220 kΩ

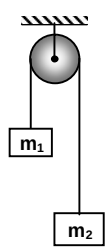
SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

21. A metallic rod of length 1 m is rigidly clamped at two points A and B as shown in the figure. Longitudinal stationary waves are setup in the rod. If the fundamental frequency of longitudinal vibration of the rod is (10 $\times$ n) kHz. Find the value of n. (Given Y = 1.6 $\times$ 10^{11} N/m^2, ρ = 2500 kg/m^3)



22. In an Atwood's machine setup as shown. Data is given by m_1 = (103.5 $\pm$ 0.2)g and m_2 = (43.5 $\pm$ 0.2)g and g = (9.80 $\pm$ 0.01) m/s^2 is acceleration due to gravity. Find the percentage error in the acceleration of the blocks.



SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

23. A small ball of mass 1 kg and charge 8 μC is placed at the centre of a uniformly charged sphere of radius 2m and charge $\frac{1}{6}$mC. A narrow smooth groove is made in the sphere from centre to the surface of sphere and the sphere is made to rotate about its vertical diameter at a constant rate of $\frac{1}{4\pi}$ revolution per second. Find the speed (in m/s) with respect to ground with which the ball slides out from the groove. Ignore any magnetic force acting on the ball. (Take $\sqrt{2} = 1.41$)

24. The eyepiece of a compound microscope has a focal length of 2.5 cm and the objective has a focal length of 1 cm. Two lenses are separated by 26 cm. Microscope is used by a person with normal eye (near point at 25 cm). If the minimum magnification of the microscope is 100x. Find the value of x.

25. A source of sound is moving with a constant speed of 20 m/s in a circle. There is a stationary observer at some distance greater than the radius of the circle from the centre of circle as shown in the fiugre. Find the difference between the maximum and minimum frequency (in Hz) observed by the observer. (Assume speed of sound to be 340 m/s and frequency of the source is 360 Hz)



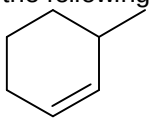
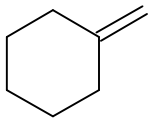
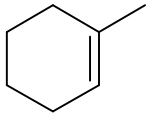
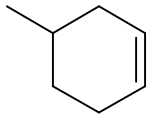
Chemistry

PART – II

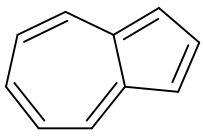
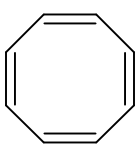
SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

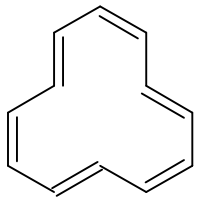
26. Which of the following is correct?
    (A) The value of steric factor P (regarding Arrheneous equation) decreases as the complexity of reacting molecules decreases.
    (B) During any individual reactive collision, the total energy of the colliding particles remain constant.
    (C) During a free radical chain reaction, if chain termination is faster than chain branching and chain propagation, explosion occurs.
    (D) In a termolecular elementary process : $O + O_2 + N_2 \longrightarrow O_3 + N_2$; N_2 serves as catalyst, while the order of reaction is two (2).

27. Which of the following is most stable?
    (A) 
    (B) 
    (C) 
    (D) 

28. In which of the following M – F bond is strongest, among the given four compounds? (M = central atom)
    (A) CF_4
    (B) NF_3
    (C) BF_3
    (D) OF_2

29. Which of the following is aromatic?
    (A) 
    (B) 

(C)



(D) A & B both

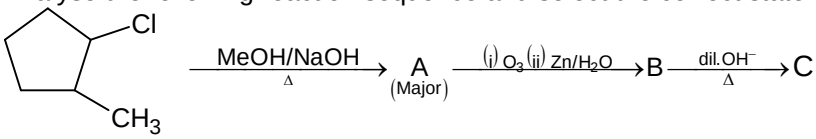
30. During extraction of sodium by Down's cell process $CaCl_2$ is mixed with NaCl, which lowers the fusion temperature, the main reason(s) for lowering the fusion temperature is:
(A) It is difficult to maintain 1081 K (fusion temperature of NaCl) during electrolysis and also at lower temperature the consumption of electricity is reduced.
(B) Both Na and Cl_2 exert corrosive nature/action on the vessel used for electrolysis at such high temperature (1081 K).
(C) At lower/reduced fusion temperature (850 K), the sodium is obtained at iron cathode in solid state.
(D) Both A and B are some of the reasons.

31. What is the pH of a saturated aqueous solution of AgOH? [Given $K_{SP}(AgOH) = 1.5 \times 10^{-8}$, log (1.225) = 0.088; log(1.5) = 0.176];
(A) 8.176
(B) 7.824
(C) 4.088
(D) 10.088

32. Which of the following is correct?
(A) Kjeldahal's method of estimation of nitrogen is not applicable to the compounds containing nitrogen in nitro groups, azo groups and nitrogen present in the ring.
(B) In Duma's method of estimation of nitrogen, the organic compound containing nitrogen is heated with Cu_2O in an atmosphere of CO_2, to yield H_2O, CO_2 and free N_2 gas along with free copper.
(C) During Carius method of estimation of halogen, a known mass of organic compound is heated with dil. HNO_3, in presence of $AgNO_3$.
(D) Phosphorous can not be detected by Lassaigne's test.

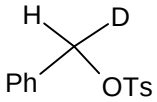
33. Which of the following is not the constituent of 'Acid Rain'?
(A) H_2CO_3
(B) H_2SO_4
(C) HNO_3
(D) H_3PO_4

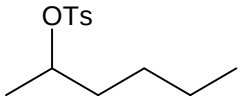
34. Analyse the following reaction sequence and select the correct statement, following the reaction?

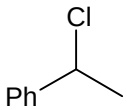


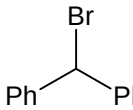
(A) 'A' has 1-degree of unsaturation.
(B) 'B' has 2-degree of unsaturation.
(C) 'C' is an a α, β-unsaturated ketone with 2-degree of unsaturation.
(D) All other three statements are incorrect.

35. In which of the following maximum stereochemical inversion is expected to occur?

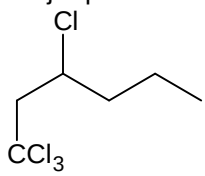
(A)  $+ \text{HCOOH} \longrightarrow S_N - \text{product}$

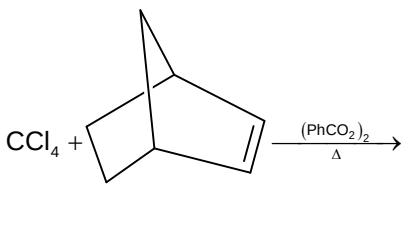
(B)  $+ CH_3COO^-NH_4^+ \longrightarrow S_N - \text{product}$

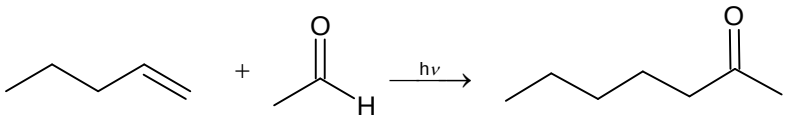
(C)  $+ CH_3COOK \xrightarrow[CH_3COOH]{} S_N - \text{product}$

(D)  $+ \text{dil. NaOH} \longrightarrow S_N - \text{product}$

36. In which of the following reaction, the given product is the major product of that reaction?

(A) $CCl_4 + CH_2 = CH - CH_2 - CH_2 - CH_3 \xrightarrow[\Delta]{(PhCO_2)_2}$  ;

(B)  ;

(C) 

(D) In all the reactions, the given products are major products.

37. During extraction process of Sn which of the following is the common method of reduction?

(A) Self reduction
(B) Aluminium reduction
(C) Electrolytic reduction
(D) Carbon reduction

38. Which of the following show geometrical isomerism as well as ionization isomerism, and possesses spin only magnetic moment of 3.87 BM?

(A) $Co(NH_3)_4Br_2Cl$
(B) $Cr(CN)(CO)_4(NO_3)Cl$
(C) $Pd(NO_2)_2(NH_3)_2$
(D) $Co(CO)_5Br(SO_4)$

39. Which of the following give Glycine?

(A) $CH_3CH_2CH=CH-CH_2-CH_3 \xrightarrow{O_3/H_2O} \xrightarrow[(ii)\,H_2O]{(i)\,Br_2/red\ P} \xrightarrow{NH_3(excess)} \xrightarrow{H_2O} ?$

(B) $CH_3CH=CHCH_3 \xrightarrow{O_3/Zn} \xrightarrow[NaCN]{NH_4Cl} \xrightarrow{H_3O^+/\Delta} \xrightarrow{Very\ dil.OH^-} ?$

(C) $CH_3CH=CHCH_3 \xrightarrow{O_3/H_2O} \xrightarrow{Br_2/red\ P} \xrightarrow{NH_3(excess)} \xrightarrow[H_2O]{} ?$

(D) $CH_3\overset{O}{\overset{\|}{C}}-COOH + NH_3 \longrightarrow \xrightarrow{H_2/Pd} ?$

40. Which one of the following statements is correct?
(Please ignore the balancing wherever is applicable)

(A) $Al + dil.\ H_2SO_4 \longrightarrow Al_2(SO_4)_3 + SO_2 + H_2O$.

(B) $Al + AlCl_3 \xrightarrow{High\ temperature} Al_2Cl_3$.

(C) $FeSO_4.(NH_4)_2\ SO_4.6H_2O$ is called Mohr's salt.

(D) AlCl is stable up to 800^oC.

41. Which of the following molecules is non linear?

(A) HCN

(B) CO_2

(C) $[AgCl_2]^-$

(D) $BaCl_2$

42. Which of the following is expected to be most acidic solution?

(A) Dilute aq. H_3BO_3 solution.

(B) Dilute aq. CH_3COOH solution.

(C) Dilute aq. HCOOH solution.

(D) Aq. $Be(OH)_2$ soluiton.

43. Which of the following process is expected to have the largest value of equilibrium constant, at room temperature?

(A) $H_3PO_2 \rightleftharpoons H^+ + HPO_2^-$; K_1

(B) $H_3PO_3 \rightleftharpoons H^+ + H_2PO_3^-$; K_2

(C) $H_3PO_4 \rightleftharpoons H^+ + H_2PO_4^-$; K_3

(D) $H_2PO_4^- \rightleftharpoons H^+ + HPO_4^{2-}$; K_4

44. Which of the following colloid combination is correctly classified?

(A) Rain cloud – Sol

(B) Soapsuds – Aerosol

(C) Silt in water – Sol

(D) Milk of magnesia – Foam

45. Cl_2 gas is passed into an aqueous solution of NaBr and NaI, followed by the addition of $CHCl_3$. What colour of $CHCl_3$ layer will be observed, ultimately?

(A) Blue

(B) Greenish yellow

(C) Reddish brown

(D) Violet

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

46. If value of equilibrium constant for the following equilibria is 10^{+x}, then find the value of 'x'.
$2Fe^{3+} + 2I^- \rightleftharpoons 2Fe^{2+} + I_2$
Given: $E^o_{Fe^{3+}/Fe^{2+}} = 0.77$ V
$E^o_{I_2/I^-} = 0.53$ V
$\frac{2.303\,RT}{F} = 0.06$

47. 18 gm of glucose ($C_6H_{12}O_6$) is treated with sufficient amount of HIO_4, under requisite conditions. How much formaldehyde (in gm) is formed?

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

48. Excess of $NH_4HS(s)$ is heated at 300 K, in a closed container and at equilibrium the partial pressure of NH_3 is found to be 1.5 atm. Some amount of NH_3 is added to the reaction mixture so that at new equilibrium, under same conditions, the partial pressure of NH_3 becomes 2 atm, what is the ratio of total pressures before adding NH_3 and after addition of NH_3.

49. Calculate the heat of formation of $H_2O(g)$ by the following data and give its numerical value only (in kJ/mole).
$\in_{H-H} = 436$ kJ / mole; $\in_{O=O} = 497$ kJ / mole;
Average $\in_{H-O} = 492$ kJ / mole

50. A metal (At. Mass = 65) has a fcc lattice with all the atoms at the lattice points. The edge length of the unit cell is 300 pm. What is its density. (in gm/cm^3) (Given: $N_A = 6.022 \times 10^{23}$)

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

51. $(p \Rightarrow q) \vee (q \Rightarrow r)$ is
(A) a tautology
(B) a contradiction
(C) equivalent to $(\sim q \Rightarrow p) \wedge (\sim r \Rightarrow q)$
(D) equivalent to $(p \vee q) \Rightarrow r$

52. Let $S = \{(x, y) : x^2 + y^2 - 4x - 2y + 4 = 0 \ \& \ x^2 + y^2 - 2y = 0\}$, Γ = set of circles which pass through all point(s) in S. Then the number of elements in Γ with unit radius that pass through all the point(s) in S and the origin is
(A) 0
(B) 1
(C) 2
(D) 3

53. The number of solution(s) of the equation $\tan^{-1}\left(x + \frac{2}{x}\right) - \tan^{-1}\frac{4}{x} - \tan^{-1}\left(x - \frac{2}{x}\right) = 0$ is
(A) 0
(B) 1
(C) 2
(D) more than 2

54. For ΔABC, $\frac{a}{1+m^2n^2} = \frac{b}{m^2+n^2} = \frac{c}{(1-m^2)(1+n^2)}$, where $m, n \in R^+$. Given three statement(s):
(I) $\angle BAC = 2\tan^{-1}\frac{m}{n}$, (II) $\angle ABC = 2\tan^{-1}mn$ and (III) area of $\Delta ABC = \frac{mnbc}{m^2+n^2}$.
Which of the following options lists all the correct statements about ΔABC?
(A) I only
(B) II and III only
(C) I and II only
(D) I, II and III

55. A statue built on a 50m high pedestal subtends 45° at two points, A_1 and A_2, 50m apart on the ground. If the straight line joining A_1 and A_2 passes through the base of the pedestal, then what is the height of the statue (without the pedestal) (assume that the statue and pedestal are vertical)
(A) $25\sqrt{2}$
(B) 50
(C) 150
(D) 250

56. If tangents are drawn from a point P to the parabola $y^2 = 4x$ with focus (S) such that the lengths of the tangents are $\sqrt{5}$ and $\sqrt{10}$, then SP is equal to
(A) $\frac{\sqrt{5}}{2}(1+\sqrt{2})$
(B) $5\sqrt{2}$

(C) $\frac{2\sqrt{10}}{1+\sqrt{2}}$

(D) none of these

57. Let the tangents drawn to the ellipse $\frac{x^2}{4}+\frac{y^2}{9}=1$ from the endpoints (A_L and B_L) of a chord (L) passing through the point (1, 1) intersect at point C_L. As L varies C_L traces which of the following curves

(A) an ellipse
(B) a parabola
(C) a circle
(D) a straight line

58. Consider a chord to the hyperbola xy = 9 which has x-intercept 3 and y-intercept 15. The mid-point of this chord is

(A) $\left(\frac{15}{2},\frac{3}{2}\right)$

(B) $\left(\frac{3}{2},\frac{15}{2}\right)$

(C) (9, 12)

(D) (12, 9)

59. A chord of $x^2-\frac{y^2}{66}=1$ subtending 90° at the centre of the hyperbola can have the equation

(A) $5x+7y=5\sqrt{3}$

(B) $3x+7y=\sqrt{59}$

(C) $7x+2y=3\sqrt{6}$

(D) $7x+4y=\sqrt{66}$

60. If (3a + c)(2a + 3b + 6c) < 0, then the number of roots of $ax^2 + bx + c = 0$ in interval (–5, 5) is

(A) exactly zero
(B) at least one
(C) exactly two
(D) can not be determined

61. Let S be the set of points in the Argand plane satisfying $|z - 3 - 4i| \le 5$. If $z_1 \in S$ such that $|z_1 - 4 + 3i| \ge |z_1 - z_0| \ \forall\ z_0 \in S$, then $\frac{10-(z_1+\bar{z}_1)}{20-i(z_1-\bar{z}_1)}$ is equal to

(A) 1

(B) $\frac{1}{2}$

(C) $\frac{1}{5}$

(D) $\frac{1}{7}$

62. If $\vec{c}$ is directed along the internal bisector of the angle between the vectors $\vec{a} = 7\hat{i} - 4\hat{j} - 4\hat{k}$ and $\vec{b} = -2\hat{i} - \hat{j} + 2\hat{k}$, with $|\vec{c}| = 5\sqrt{6}$, then $\vec{c}$ is

(A) $\frac{5}{3}\left(\hat{i} - 7\hat{j} + 2\hat{k}\right)$

(B) $\frac{5}{3}\left(5\hat{i} + 5\hat{j} + 2\hat{k}\right)$

(C) $\frac{5}{3}\left(\hat{i} + 7\hat{j} + 2\hat{k}\right)$

(D) $\frac{5}{3}\left(-5\hat{i} + 5\hat{j} + 2\hat{k}\right)$

63. If the area of the region bounded by the curves $xy + x = 1$, $y^2 = 12x$ and the x-axis is A, then

(A) $A > 1$

(B) $A < \ln 2$

(C) $A^2 < \frac{1}{\sqrt{2}}$

(D) $A > \frac{1}{\sqrt{2}}$

64. $\lim\limits_{n\to\infty} \sum\limits_{i=1}^{n-1} \frac{n^2 i}{i^4 + n^4}$ is equal to

(A) $\ln 2$

(B) $\frac{1}{2}\ln 2$

(C) $\frac{\pi}{4}$

(D) $\frac{\pi}{8}$

65. If $(z + 1)^{2n} + (z - 1)^{2n} = 0$, where z is a complex number and $n \in N$, then which of the following is always true?

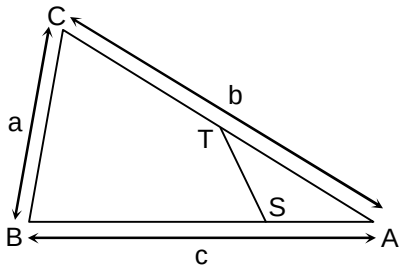
(A) $\bar{z} = z$

(B) $z = \bar{z}^2$

(C) $z + \bar{z} = 0$

(D) $z^2 + \bar{z}^2 = 1$

66. Given a triangular plot of land as per the figure, the owner wants to build a fence ST such that the plot is divided into 2 equal parts. The minimum length of fencing required is (where s is the semi-perimeter of ΔABC)



where $a \le b \le c$

(A) $\sqrt{2(s-b)(s-c)}$

(B) $\sqrt{s(s-a)}$

(C) $\sqrt{a(b+c)}$

(D) $\frac{s}{2}$

67. If $S_n = \frac{1}{1\cdot3\cdot4} + \frac{1}{2\cdot4\cdot5} + ... + \frac{1}{n(n+2)(n+3)}$, then $\lim_{n\to\infty} S_n$ is equal to

(A) $\frac{1}{2}$

(B) $\frac{5}{36}$

(C) $\frac{5}{12}$

(D) $\frac{5}{6}$

68. If A, B $\in$ R such that the function $f(x) = \begin{cases} \frac{\pi}{2} & ; \text{ if } x \in [0,4] \\ B\tan^{-1}\frac{2}{x-4} & ; \text{ if } x \in (4,6) \\ \sin^{-1}(7-x) + A\frac{\pi}{4} & ; \text{ if } x \in [6,8] \end{cases}$, is continuous in [0, 8], then $A^2 + B^2$ is equal to

(A) 1
(B) 2
(C) 3
(D) 4

69. If $f(x) = \begin{vmatrix} \cos x & x & 1 \\ 2\sin x & x^2 & 2x \\ \tan x & x & 1 \end{vmatrix}$, then $\lim_{x\to0} \frac{f'(x)}{x}$ is equal to

(A) –2
(B) 0
(C) 1
(D) 2

70. If y = f(x) satisfies the differential equation $x\frac{dy}{dx} + y(2x+1) = xe^x$ and f(0) = 0, then the number of solution(s) of $f(x) = \frac{2}{x}$ is equal to

(A) 0
(B) 1
(C) 2
(D) more than 2

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

71. 5 boys (B_1, B_2, B_3, B_4 and B_5) and 5 girls (G_1, G_2, G_3, G_4 and G_5) are to be seated around a round table such that boy and girl sit alternately and B_i does not sit beside G_i $\forall$ i $\in$ {1, 2, 3, 4, 5}. If the number of such arrangements is N, then the sum of digits of N is equal to

72. A sample of 20 observations has mean of 50 and variance of 1, while a sample of 40 observations has mean of 50 and standard deviation 2. The 2 samples are combined to give complete set of 60 observations with variance σ^2, then $3\sigma^2$ is equal to

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

73. Let $f(x) = 27x^3 + 8x^2 + x + 1$, then $\lim\limits_{x\to\infty} \dfrac{x^{\frac{1}{3}}}{f^{-1}(27x) - f^{-1}(x)}$ is equal to

74. A plane (P) is parallel to the line $L \equiv \dfrac{x-1}{1} = \dfrac{y-2}{1} = \dfrac{z-3}{1}$. If the image of origin in P lies on the line L, then the square of distance of (3, 2, 1) from P is equal to

75. An aeroplane has 100 seats (numbered 1 to 100) and 100 passengers waiting to board each having a ticket with a number from 1 to 100. No number is on 2 tickets or on 2 seats. The rules of boarding are as below:
    (i) Passengers board in the order of the number on their respective ticket,
    (ii) The first passenger to board can sit on any seat, and
    (iii) A passenger with ticket number i ($i \neq 1$) boards the plane and sits on seat numbered 'i' if it is empty, however if that seat is occupied he can sit on any empty seat.

    If p is the probability that the 100th passenger sits in his assigned seat, then $p^2 + (1 - p)^2$ is equal to